Andrea Zhang

AI/ML Engineer | Software Engineer | Distributed Systems

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Languages: Italian (native) | Mandarin Chinese (native) | English (proficient)

Computer Science and Engineering MSc graduate with hands-on experience building machine learning software, distributed systems, and full-stack applications in Python, Java, Dart, and TypeScript. Member of the 1st-place team in the ACM RecSys Challenge 2026.

EXPERIENCE

Junior Consultant - AI Automation | Reply, Milan

Mar - Sep 2025 (6 months)

Built a Microsoft Copilot Studio workflow for an enterprise chatbot that interprets user queries and retrieves relevant Excel and PowerPoint files, translating a business need into a working AI-assisted process.

SELECTED AI/ML AND SOFTWARE PROJECTS

1st Place - ACM RecSys Challenge 2026

Analyzed 388,776 conversational turns, designed session-aware validation, and evaluated collaborative filtering, two-tower, and graph-based recommenders. The final team system ranked 1st on Blind-B (0.6185 nDCG@20); co-authored the resulting ACM publication.

Reward learning from human feedback - Master's thesis

Built a modular Python framework combining demonstrations and preferences to train reward models for simulated driving, with reproducible experiments and reliability-weighted gradient fusion.

High-performance streaming regression in CapyMOA

Implemented a Java/MOA Hoeffding Adaptive Tree Regressor exposed through Python. Matched River's deterministic predictions while running 3.4-6.0x faster across eight data streams.

Tripline - full-stack collaborative travel application

Built a Flutter/Dart product with MVVM and Riverpod, backed by Firebase and TypeScript Cloud Functions. Implemented realtime collaboration, role-based access, notifications, external APIs, and broad unit and integration test coverage.

Codex Naturalis - multiplayer client-server application

Built a Java adaptation of the board game with RMI and socket transports, GUI and TUI clients, multiple concurrent games, disconnection resilience, chat, and JUnit tests. Achieved 96% line coverage across the model package.

EDUCATION

Politecnico di Milano - MSc, Computer Science and Engineering

Sep 2024 - Oct 2026

Graduated 21 Oct 2026 with 110/110 cum laude. Coursework: Distributed Systems, Machine Learning, Recommender Systems, Artificial Neural Networks and Deep Learning.

Politecnico di Milano - BSc, Engineering of Computing Systems

Sep 2021 - Sep 2024

Final grade: 107/110. Best Freshmen Award, engineering class L-8 (Fondo Giovani).

TECHNICAL SKILLS

AI & data: machine learning, recommender systems, reinforcement learning, PyTorch, TensorFlow/Keras, Stable-Baselines3, MOA/CapyMOA | Programming: Python, Java, Dart, TypeScript, HTML, CSS | Engineering: distributed systems, TCP/UDP networking, Git, Maven, Firebase, software testing, performance evaluation